OPENNESS OF THE FLAT LOCUS OF A FINITELY PRESENTED MODULE OVER AN ARBITRARY BASE

THE STACKS PROJECT AUTHORS

Editorial note. Public1510 assumes locally finitely many weakly associated base points and a finite-type sheaf; it uses the usual finitely-presented openness theorem as input. This task proves the latter for arbitrary base rings. Neither hypothesis set is merely the same renamed assertion.

Editorial note. The source describes the localized polynomial fibre global dimension as n; the needed bound is at most n, which suffices for the nth syzygy. One may add dummy polynomial variables to avoid the harmless n=0 indexing convention.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Finite-presentation approximation and descent of flatness, Tor exactness, polynomial-ring global dimension, local flat/free criterion and finite free resolutions. The local fibrewise-exactness and final cokernel arguments are retained as author proof, not removed.

Source reference. Original lemma 00ME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 820–820. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 00N1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 843–843. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00RA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 996–996. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00R1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 987–987. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:algebra.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00QX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 983–983. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00R6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 992–992. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00LP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 795–795. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00HM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 648–648. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Theorem 0.1. *Let R be a ring. Let $R \to S$ be a ring map of finite presentation. Let M be a finitely presented S-module. The set*

$$\{\mathfrak{q} \in \mathrm{Spec}(S) \mid M_{\mathfrak{q}} \text{ is flat over } R\}$$

is open in $\mathrm{Spec}(S)$.

Lemma 0.2. *Let $R \to S$ be a local homomorphism of Noetherian local rings. Let $\mathfrak{m}$ be the maximal ideal of R. Let M be a nonzero finite S-module. Suppose that (a) $M/\mathfrak{m}M$ is a free $S/\mathfrak{m}S$-module, and (b) M is flat over R. Then M is free and S is flat over R.*

Proof. Let $\overline{x}_1, \ldots, \overline{x}_n$ be a basis for the free module $M/\mathfrak{m}M$. Choose $x_1, \ldots, x_n \in M$ with x_i mapping to $\overline{x}_i$. Let $u : S^{\oplus n} \to M$ be the map which maps the ith standard basis vector to x_i. By Lemma 00ME we see that u is injective. On the other hand, by Nakayama's Lemma 00DV the map is surjective. The lemma follows. □

Lemma 0.3. *Let $R \to S$ be a local homomorphism of local Noetherian rings. Let $\mathfrak{m}$ be the maximal ideal of R. Let $0 \to F_e \to F_{e-1} \to \ldots \to F_0$ be a finite complex of finite S-modules. Assume that each F_i is R-flat, and that the complex $0 \to F_e/\mathfrak{m}F_e \to F_{e-1}/\mathfrak{m}F_{e-1} \to \ldots \to F_0/\mathfrak{m}F_0$ is exact. Then $0 \to F_e \to F_{e-1} \to \ldots \to F_0$ is exact, and moreover the module* $\mathrm{Coker}(F_1 \to F_0)$ *is R-flat.*

Proof. By induction on e. If $e = 1$, then this is exactly Lemma 00ME. If $e > 1$, we see by Lemma 00ME that $F_e \to F_{e-1}$ is injective and that $C = \mathrm{Coker}(F_e \to F_{e-1})$ is a finite S-module flat over R. Hence we can apply the induction hypothesis to the complex $0 \to C \to F_{e-2} \to \ldots \to F_0$. We deduce that $C \to F_{e-2}$ is injective

and the exactness of the complex follows, as well as the flatness of the cokernel of $F_1 \to F_0$. □

Lemma 0.4. *Let $R \to S$ be a ring map. Consider a finite homological complex of finite free S-modules:*

$$F_\bullet : 0 \to S^{n_e} \xrightarrow{\varphi_e} S^{n_{e-1}} \xrightarrow{\varphi_{e-1}} \ldots \xrightarrow{\varphi_{i+1}} S^{n_i} \xrightarrow{\varphi_i} S^{n_{i-1}} \xrightarrow{\varphi_{i-1}} \ldots \xrightarrow{\varphi_1} S^{n_0}$$

For every prime $\mathfrak{q}$ of S consider the complex $\overline{F}_{\bullet,\mathfrak{q}} = F_{\bullet,\mathfrak{q}} \otimes_R \kappa(\mathfrak{p})$ where $\mathfrak{p}$ is inverse image of $\mathfrak{q}$ in R. Assume R is Noetherian and there exists an integer d such that $R \to S$ is finite type, flat with fibres $S \otimes_R \kappa(\mathfrak{p})$ Cohen-Macaulay of dimension d. The set

$$\{\mathfrak{q} \in \mathrm{Spec}(S) \mid \overline{F}_{\bullet,\mathfrak{q}} \text{ is exact}\}$$

is open in $\mathrm{Spec}(S)$.

Proof. Let $\mathfrak{q}$ be an element of the set defined in the lemma. We are going to use Proposition 00N1 to show there exists a $g \in S$, $g \notin \mathfrak{q}$ such that $D(g)$ is contained in the set defined in the lemma. In other words, we are going to show that after replacing S by S_g, the set of the lemma is all of $\mathrm{Spec}(S)$. Thus during the proof we will, finitely often, replace S by such a localization. Recall that Proposition 00N1 characterizes exactness of complexes in terms of ranks of the maps φ_i and the ideals $I(\varphi_i)$, in case the ring is local. We first address the rank condition. Set $r_i = n_i - n_{i+1} + \ldots + (-1)^{e-i} n_e$. Note that $r_i + r_{i+1} = n_i$ and note that r_i is the expected rank of φ_i (in the exact case).

By Lemma 0.3 we see that if $\overline{F}_{\bullet,\mathfrak{q}}$ is exact, then the localization $F_{\bullet,\mathfrak{q}}$ is exact. In particular the complex $F_\bullet$ becomes exact after localizing by an element $g \in S$, $g \notin \mathfrak{q}$. In this case Proposition 00N1 applied to all localizations of S at prime ideals implies that all $(r_i+1) \times (r_i+1)$-minors of φ_i are zero. Thus we see that the rank of φ_i is at most r_i.

Let $I_i \subset S$ denote the ideal generated by the $r_i \times r_i$-minors of the matrix of φ_i. By Proposition 00N1 the complex $\overline{F}_{\bullet,\mathfrak{q}}$ is exact if and only if for every $1 \leq i \leq e$ we have either $(I_i)_\mathfrak{q} = S_\mathfrak{q}$ or $(I_i)_\mathfrak{q}$ contains a $S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}$-regular sequence of length i. Namely, by our choice of r_i above and by the bound on the ranks of the φ_i this is the only way the conditions of Proposition 00N1 can be satisfied.

If $(I_i)_\mathfrak{q} = S_\mathfrak{q}$, then after localizing S at some element $g \notin \mathfrak{q}$ we may assume that $I_i = S$. Clearly, this is an open condition.

If $(I_i)_\mathfrak{q} \neq S_\mathfrak{q}$, then we have a sequence $f_1, \ldots, f_i \in (I_i)_\mathfrak{q}$ which form a regular sequence in $S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}$. Note that for any prime $\mathfrak{q}' \subset S$ such that $(f_1, \ldots, f_i) \not\subset \mathfrak{q}'$ we have $(I_i)_{\mathfrak{q}'} = S_{\mathfrak{q}'}$. Thus the result follows from Lemma 00RA. □

Proof. Let $\mathfrak{q} \in \mathrm{Spec}(S)$ be a prime. Let $\mathfrak{p} \subset R$ be the inverse image of $\mathfrak{q}$ in R. Note that $M_\mathfrak{q}$ is flat over R if and only if it is flat over $R_\mathfrak{p}$. Let us assume that $M_\mathfrak{q}$ is flat over R. We claim that there exists a $g \in S$, $g \notin \mathfrak{q}$ such that M_g is flat over R.

We first reduce to the case where R and S are of finite type over $\mathbf{Z}$. Choose a directed set Λ and a system $(R_\lambda \to S_\lambda, M_\lambda)$ as in Lemma 00R1. Set $\mathfrak{p}_\lambda$ equal to the inverse image of $\mathfrak{p}$ in R_λ. Set $\mathfrak{q}_\lambda$ equal to the inverse image of $\mathfrak{q}$ in S_λ. Then the system

$$((R_\lambda)_{\mathfrak{p}_\lambda}, (S_\lambda)_{\mathfrak{q}_\lambda}, (M_\lambda)_{\mathfrak{q}_\lambda})$$

is a system as in Lemma 00QX. Hence by Lemma 00R6 we see that for some λ the module M_λ is flat over R_λ at the prime $\mathfrak{q}_\lambda$. Suppose we can prove our claim for the system $(R_\lambda \to S_\lambda, M_\lambda, \mathfrak{q}_\lambda)$. In other words, suppose that we can find a $g \in S_\lambda$, $g \notin \mathfrak{q}_\lambda$ such that $(M_\lambda)_g$ is flat over R_λ. By Lemma 00R1 we have $M = M_\lambda \otimes_{R_\lambda} R$

and hence also $M_g = (M_\lambda)_g \otimes_{R_\lambda} R$. Thus by Lemma 00HI we deduce the claim for the system $(R \to S, M, \mathfrak{q})$.

At this point we may assume that R and S are of finite type over $\mathbf{Z}$. We may write S as a quotient of a polynomial ring $R[x_1, \ldots, x_n]$. Of course, we may replace S by $R[x_1, \ldots, x_n]$ and assume that S is a polynomial ring over R. In particular we see that $R \to S$ is flat and all fibres rings $S \otimes_R \kappa(\mathfrak{p})$ have global dimension n.

Choose a resolution $F_\bullet$ of M over S with each F_i finite free, see Lemma 00LP. Let $K_n = \mathrm{Ker}(F_{n-1} \to F_{n-2})$. Note that $(K_n)_\mathfrak{q}$ is flat over R, since each F_i is flat over R and by assumption on M, see Lemma 00HM. In addition, the sequence

$$0 \to K_n/\mathfrak{p}K_n \to F_{n-1}/\mathfrak{p}F_{n-1} \to \ldots \to F_0/\mathfrak{p}F_0 \to M/\mathfrak{p}M \to 0$$

is exact upon localizing at $\mathfrak{q}$, because of vanishing of $\mathrm{Tor}_i^{R_\mathfrak{p}}(\kappa(\mathfrak{p}), M_\mathfrak{q})$. Since the global dimension of $S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}$ is n we conclude that $K_n/\mathfrak{p}K_n$ localized at $\mathfrak{q}$ is a finite free module over $S_\mathfrak{q}/\mathfrak{p}S_\mathfrak{q}$. By Lemma 0.2 $(K_n)_\mathfrak{q}$ is free over $S_\mathfrak{q}$. In particular, there exists a $g \in S$, $g \notin \mathfrak{q}$ such that $(K_n)_g$ is finite free over S_g.

By Lemma 0.4 there exists a further localization S_g such that the complex

$$0 \to K_n \to F_{n-1} \to \ldots \to F_0$$

is exact on *all fibres* of $R \to S$. By Lemma 0.3 this implies that the cokernel of $F_1 \to F_0$ is flat. This proves the theorem in the Noetherian case. □

References